UGH/INT/31/07/26

RollNo:.....

# MARAUGUSTHINOSECOLLEGERAMAPURAM

## MGU-BCA(H) INTERNAL EXAMINATION,JULY 2026

## MG3CCRBCA203–DESIGN AND ANALYSIS OF ALGORITHMS

### Third Semester

**Time: 1 Hour** **MaximumMarks:35**

**(Remember (K), Understand (U), Apply (A), Analyse (An))**

***Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.***

### PartA

***(Answer all questions. Each Question carries 2 Marks).***

1. Define algorithm. [K][1]
2. Explain Recursion. [A][1]
3. Explain Big Oh Notation. [A][1]
4. Explain best, worst and average case complexity. [A][1]
5. What is the complexity of binary search algorithm? [U][2]

(5 X 2=10)

### *PART B*

***(Answer any THREE questions. Each question carries 5 marks)***

6. Explain Different Algorithm Design Techniques. [A][1]
7. Explain Space And Time Complexity of an Algorithm. [A][1]
8. Explain Asymptotic Notation. [A][1]
9. Discuss Control abstraction for Divide and Conquer Method. [An][2]

(3X 5=15)

### *PART C*

***(Answer any One question. Each question carries 10 marks)***

10. Explain Merge Sort Algorithm with an Example. [A][2]
11. Explain the algorithm for finding maximum and minimum in an array with Example. [A][2]

(1 X 10= 10)

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